

On the Images of Certain Projections

In lecture on Monday, Feb. 26, 2018, we saw the following theorems:¹

Theorem A (Paraphrase of Exercise 2.24, Page 29, of Harris) *For any projective variety $X \subseteq \mathbb{P}_K^m$ and any regular map $f : X \rightarrow \mathbb{P}_K^n$, the graph $\Gamma_f := \{(x, f(x)) \mid x \in X\}$ is a Zariski-closed subset of $\mathbb{P}_K^m \times \mathbb{P}_K^n$. ■*

Theorem B (Theorem 3.12, Page 38, of Harris) *Let $Y \subseteq K^m$ be any Zariski-closed set, X any Zariski-closed subset of $Y \times \mathbb{P}_K^n$, and $\pi : Y \times \mathbb{P}_K^n \rightarrow Y$ the projection defined by $\pi((y_1, \dots, y_m) \times [p_0 : \dots : p_n]) := (y_1, \dots, y_m)$. Then $\pi(X)$ is Zariski-closed in Y (and thus in K^m as well).*

Theorem C *Let $\Delta_n := \text{Conv}\{\mathbf{O}, e_1, \dots, e_n\}$ denote the standard n -simplex in $\mathbb{R}^n$. Then, given any positive integers $d_1, \dots, d_{n+1}$, there is a polynomial $\text{Res}_{(d_1, \dots, d_{n+1})}$ in the ring $\mathbb{Z}[c_{i,a} \mid (i, a) \in \{1, \dots, n+1\} \times (d_i \Delta_n) \cap \mathbb{Z}^n]$ with the following property:*

$\text{Res}_{(d_1, \dots, d_{n+1})}(c) = 0$ for some coefficient vector $c \in K^{(d_1+n) + \dots + (d_{n+1}+n)}$
 $\iff$ the system of equations $F := (f_1, \dots, f_{n+1})$, with

$$\begin{aligned} f_1(x) &:= \sum_{a \in d_1 \Delta_n \cap \mathbb{Z}^n} c_{1,a} x^a \\ &\vdots \\ f_{n+1}(x) &:= \sum_{a \in d_{n+1} \Delta_n \cap \mathbb{Z}^n} c_{n+1,a} x^a, \end{aligned}$$

(and $c = (c_{i,a} \mid (i, a) \in \{1, \dots, n+1\} \times (d_i \Delta_n) \cap \mathbb{Z}^n)$) has a root in $\mathbb{P}_K^n$. In particular, for each i , $\text{Res}_{(d_1, \dots, d_{n+1})}$ is homogeneous of degree $\frac{\prod_{j=1}^{n+1} d_j}{d_i}$ with respect to $(c_{i,a} \mid a \in d_i \Delta_n)$. ■

Remark 1 While we could just use the simpler “ 2×1 ” Sylvester resultant to prove Theorem B, this would restrict the kinds of projections we’d like to use. Also, if one cares about explicit bounds on the complexity of the image (such as degree bounds or coefficient height bounds for the defining polynomials of the image), then composed Sylvester resultants yield upper bounds that are magnitudes away from optimal. So while the multivariate resultant $\text{Res}_{(d_1, \dots, d_{n+1})}$ no longer admits an easy determinantal formula for $n \geq 2$, it definitely has other pay-offs for $n \geq 2$. ◇

Remark 2 The multivariate resultant satisfies many other important properties that we’ll not pursue here. For instance, $\text{Res}_{(d_1, \dots, d_{n+1})}$ is an irreducible polynomial, and also satisfies some functoriality properties with respect to replacing an f_i by a polynomial linear combination. We’ll prove the irreducibility statement later in class. ◇

These theorems were preparation for proving the following central theorem on maps between projective varieties:

Theorem 3 *The image of any projective variety under a regular map is a projective variety.*

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¹ Please note that we are assuming that K is an algebraically closed field.

In lecture, we saw that Theorem A was a variant of a homework exercise, and that Theorems A and B together easily imply Theorem 3. However, we omitted the proof of Theorems B and C. Here we will prove Theorems B and C.

Note: For the sake of posting this draft quickly, I will fill in the proof of Theorem C later, but again point out that Ioannis Emiris' Ph.D. thesis is an excellent source for the refinement of Theorem C to sparse resultants.

A final result that will help simplify our proof is the following:

Theorem D Suppose $Z \subsetneq \mathbb{P}_K^n$ is Zariski-closed and d -dimensional with $d < n$. Then there is an $(n - d - 1)$ -dimensional subspace H with $Z \cap H = \emptyset$. ■

Theorem D is not hard to prove from the basic properties of cones over varieties. The H that do intersect Z turn out to be rare: They in fact form a proper, closed subvariety of a suitable (affine) Grassmannian known as a *Chow variety*. More on this later...

Proof of Theorem B: First note that we may reduce to the special case where X is irreducible, since an arbitrary X has only finitely many irreducible components, and closed sets are closed under finite unions. Next, note also that any Zariski-closed subset of $K^m \times \mathbb{P}_K^n$ has closed intersection with $Y \times \mathbb{P}_K^n$, so we can reduce to the special case $Y = K^m$.

Now let $I \subseteq K[x_1, \dots, x_m, y_0, \dots, y_n]$ denote the defining ideal of X . Note that I is generated by polynomials homogeneous in $y_0, \dots, y_n$. If I is principal then the closedness of $\pi(X)$ follows easily by a small variation of the solution to Problem #2 of HW#5. So we may now assume that I is minimally generated by $f_1, \dots, f_r$ with $r \geq 2$, and X has dimension $d \leq m + n - 2$. By the dimension theorems from Section 6 of Chapter 1 of our text, we must also have $d \geq m + n - r$. We now show how to reduce to the special case where $n = 1$. The main trick will be to eliminate $j - 1$ variables (for some $j \in \{2, \dots, \min\{r, n + 1, m + n - d + 1\}\}$) in order to strictly decrease n .

In particular, consider the intermediate projection $\rho_j : K^m \times \mathbb{P}_K^n \dashrightarrow K^m \times \mathbb{P}_K^{n-j+1}$ defined by $\rho_j((x_1, \dots, x_m) \times [y_0 : \dots : y_n]) := (x_1, \dots, x_m) \times [y_0 : \dots : y_{n-j+1}]$: This projection is not a regular map, but ρ_j is regular away from the $(j - 2)$ -plane H defined by $y_0 = \dots = y_{n-j+1} = 0$. Note also that the image of π is unaffected by the natural action of $GL_{n+1}(K)$ on $\mathbb{P}_K^n$. We will use ρ_j , along with $GL_{n+1}(K)$ -action and the multivariate resultant, to reduce n and complete our proof.

Toward this end, set $j := \min\{r, n + 1, m + n - d + 1\}$ and observe that $j \geq 2$ and $(j - 2) + d \leq m + n - 1$ by construction. So by $GL_{n+1}(K)$ -action, we may assume that $X \cap (K^m \times H) = \emptyset$ and thus have ρ_j regular on X .

Now, assume temporarily that $r \leq n + 1$. For any i , let d_i denote the degree of f_i with respect to the variables $(y_0, \dots, y_n)$. Considering $f_1, \dots, f_j$ as polynomials in the ring $K[x_1, \dots, x_m, y_0, \dots, y_{n-j+1}][y_{n-j+2}, \dots, y_n]$, it is then easily checked that $\text{Res}_{(d_1, \dots, d_j)}(f_1, \dots, f_j)$ (applied to eliminate the variables $y_{n-j+2}, \dots, y_n$) is homogeneous in $y_0, \dots, y_{n-j+1}$. In particular, thanks to Theorem C, $X' := \rho_j(X)$ is exactly the hypersurface in $K^m \times \mathbb{P}_K^{n-j+1}$ defined by the zero set of $\text{Res}_{(d_1, \dots, d_j)}(f_1, \dots, f_j) \in K[x_1, \dots, x_m, y_0, \dots, y_{n-j+1}]$. So X' is Zariski-closed, and we merely need to prove that the projection $\pi' : K^m \times \mathbb{P}_K^{n-j+1} \rightarrow K^m$ has $\pi'(X')$ Zariski-closed. So we have succeeded in reducing to a case of strictly lower dimension, as promised.

To address the case where $r > n + 1$, we simply repeat the last argument, but with a slightly more complicated description of X' as a Zariski-closed subset of $K^m \times \mathbb{P}_K^{n-j+1}$: Thanks to Theorem C, $X' := \rho_j(X)$ is exactly the intersection of the hypersurfaces defined by the zero sets of $\text{Res}_{(d_1, \dots, d_j)}(f_{i_1}, \dots, f_{i_j}) \in K[x_1, \dots, x_m, y_0, \dots, y_{n-j+1}]$, as $\{i_1, \dots, i_j\}$ ranges over all cardinality j subsets of $\{1, \dots, r\}$. So, again, X' is Zariski-closed, and we merely need to

prove that the projection $\pi' : K^m \times \mathbb{P}_K^{n-j+1} \longrightarrow K^m$ has $\pi'(X')$ Zariski-closed. So we have again successfully reduced to a case of strictly lower dimension.

To conclude, we need only address the case $n = 1$. But this case is already implicit in our preceding arguments: The $r = 1$ case is the case where I is principal, which we already proved earlier. The cases $r \geq 2$ (already addressed) show that $\pi(X)$ is a Zariski-closed subset of $K^m \times \mathbb{P}_K^0$ when $n = 1$. In particular, while the defining polynomials we obtain via resultants are homogeneous in y_0 (possibly of positive degree), $\mathbb{P}_K^0$ consists of the single point $[1]$. So we can set $y_0 = 1$ to obtain defining polynomials in K^m , and be done. ■